OFFICIAL EXAM BLUEPRINT PREPARATION GUIDE

Panaversity PCAO-F & Anthropic CCAO-F

Chapter: Code_You_Never_Write

40 High-Yield Scenario-Based Examination Items • Cut Score: 720 / 1,000 (72%) • Psychometric Standard

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Examination Instructions: Part 1 contains 40 numbered scenario-based items with zero answers or spoilers. Complete Part 1 under simulated exam conditions (target time: 60 minutes). Then consult Part 2 for the official answer keys, direct source citations, and comprehensive distractor dissections.

PART 1: QUESTIONS ONLY (SPOILER-FREE SIMULATION)

QUESTION 1 — Domain 1: AI Foundations & Architecture • The Client Mindset vs Writing Code Syntax

A regional bakery chain operations director needs an automated spreadsheet comparing flour spot prices across six regional mills. A junior analyst argues that asking an AI assistant to generate prose recommendations is identical to asking it to write a self-executing Python script. Why is this distinction critical between generating code and generating text?

A) Modern relational healthcare databases automatically eliminate all scheduling mismatches without human review.
B) Professionals now act as clients who describe problems clearly, while AI writes, executes, and repairs the code.
C) Programming languages like Python are being permanently deprecated by major cloud hosting operating systems.
D) Bootcamp curricula are legally prohibited from teaching automated medical record reconciliation algorithms.

QUESTION 2 — Domain 1: AI Foundations & Architecture • Code Exactness in Both Directions

A network administrator instructs an AI assistant to write and execute a script to rename 300 router configuration files. Due to an unstated date formatting ambiguity in the prompt, the script renames every single configuration file to an identical generic filename, overwriting the entire directory in seconds. What core property of code does this incident illustrate?

A) Code executes instructions with flawless mechanical precision, carrying out flawed briefs just as ruthlessly.
B) The model's underlying neural network experienced catastrophic memory corruption during file token encoding.
C) The server's local operating system automatically corrupts text files when batch rename scripts exceed 100 rows.
D) The network administrator failed to compile the Python script into raw hexadecimal assembly code before running.

QUESTION 3 Domain 1: AI Foundations & Architecture • Automated Language Selection: Output Governs Choice

A high school registrar asks an AI assistant to build two distinct deliverables: (1) a monthly script that ingests student attendance logs and totals absences by homeroom, and (2) an interactive web page with clickable buttons that allows teachers to calculate grade averages in their browser. Under standard delegated computing principles, how should the registrar select the programming languages?

A) The registrar should not choose; the AI automatically selects Python for data totals and JavaScript for web UI.
B) The registrar must demand C++ for tabular data processing and SQL database triggers for clickable web interfaces.
C) The registrar must specify Python for both tasks to ensure identical algorithmic execution speeds in the browser.
D) The registrar must mandate Rust for all calculations to comply with federal student privacy security mandates.

QUESTION 4 Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • The VPRF Framework: Mind Work vs Hands Work

A corporate executive assistant reviews four requests: (1) 'Draft a diplomatic email declining a vendor dinner,' (2) 'Summarize key themes across this single 2-page memo,' (3) 'Total 4,500 expense rows across twelve spreadsheets,' and (4) 'Evaluate whether our company slogan sounds outdated.' Which request unequivocally crosses the dividing line into a code problem?

A) Drafting the diplomatic email, because business correspondence requires zero-defect grammatical precision rules.
B) Evaluating the company slogan, because brand marketing assessments require statistical sentiment analysis models.
C) Totalling the 4,500 expense rows, because it fires the Volume, Precision, and Files criteria of the VPRF filter.
D) Summarizing the 2-page memo, because multi-paragraph document reading mandates internal Python script execution.

QUESTION 5 Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • The Precision Signal in Casual Conversational Clothing

A university admissions officer asks an AI chat assistant: 'Roughly how many applicants from our engineering pool have unpaid application fees?' The assistant skims the attached applicant table and pleasantly reports that exactly eight applicants owe fees. Later audits reveal eleven applicants had unpaid balances. What fundamental mistake did the officer make?

A) Using a foundation model that lacked pre-trained knowledge regarding higher education tuition billing software.
B) Prompting the assistant on a weekend when commercial cloud inference data centers run at reduced compute capacity.
C) Failing to convert the applicant spreadsheet into encrypted vector embeddings before querying the chat interface.
D) Asking conversationally for an estimate on a task that fired Precision, allowing the AI to glance instead of compute.

QUESTION 6 **Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • The Repetition Signal: Converting Chores into Assets**

A freight dispatcher writes a brief for an automated script to calculate vehicle detention fees across regional depots. Which question belongs under the specification of what the script must produce?

A) It transforms a recurring manual chore into a permanent, reusable script asset that executes in seconds forever.
B) It enables the assistant to automatically register a corporate copyright on the generated Python sorting script.
C) It permanently rewrites the underlying foundation model neural weights to automate freight scheduling globally.
D) It forces cloud model providers to waive all enterprise inference API token consumption fees for that workflow.

QUESTION 7 **Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • Misapplying Code to Qualitative Answer Problems**

A marketing director prompts an AI model: 'Write and run Python code to analyze whether this candidate's cover letter exhibits authentic leadership empathy and cultural alignment.' The assistant generates a fragile script that counts occurrences of the word 'team' and outputs a flawed score. What error was committed?

A) Failing to attach the candidate's criminal background check records to the cover letter evaluation prompt context.
B) Forcing programmatic code execution on a subjective qualitative problem that required conversational AI reasoning.
C) Selecting Python instead of JavaScript to parse candidate leadership communication patterns in text files.
D) Neglecting to configure an automated Kubernetes server cluster to compile the text sentiment analysis script.

QUESTION 8 **Domain 2: Prompt Engineering & In-Context Learning • The Three-Line Mandate: The Cheapest Lie Detector**

A financial comptroller uploads an unfamiliar 8,000-row general ledger export and needs an audit of total travel expenditures. To guarantee that the AI executes deterministic programmatic calculation rather than plausible statistical estimation, which prompt structure should she mandate?

A) Paste the first ten rows of the ledger into the prompt and ask the model to visually extrapolate the annual totals.
B) Instruct the model to execute at temperature zero while translating all financial debit entries into Latin phrases.
C) Request that the assistant adopt the persona of a senior forensic accountant and swear that all figures are true.
D) Demand code execution, require the code block to be displayed, and ask for row counts and column names first.

QUESTION 9 **Domain 4: Model Evaluation, Validation & Error Mitigation • Verifying Code Proof: The Tell**

A payroll manager uploads a monthly timesheet CSV and asks the assistant: 'Total the overtime hours for Department B.' The assistant instantly replies with an attractive formatted summary stating that Department B accumulated exactly 342.5 overtime hours. No code block appears in the response. How should the manager treat this output?

A) As an authoritative certified payroll calculation because foundation models execute background code silently.
B) As an unverified opinion about the data that must not be trusted until code execution is proven and inspected.
C) As a legal labor violation because AI assistants are strictly prohibited from generating employee overtime reports.
D) As an encrypted data packet that requires specialized cryptographic keys to display the underlying calculations.

QUESTION 10 **Domain 2: Prompt Engineering & In-Context Learning • Refraining from Specifying Programming Languages**

A junior data coordinator writes a prompt: 'Write and run a script in Julia version 1.8 to calculate the average customer churn rate from this CSV.' Why is it operationally superior to instruct the AI to compute, without mandating a specific programming language?

A) Foundation models experience catastrophic loss of natural language understanding when instructed to write Julia.
B) Commercial cloud sandboxes automatically terminate user chat sessions if any language other than Python is named.
C) Forcing a specific language micromanages a technical choice that the AI is better equipped to determine automatically.
D) Naming a programming language triggers proprietary enterprise licensing fees payable to the language creator.

QUESTION 11 **Domain 2: Prompt Engineering & In-Context Learning • The Diagnostic Interrogation: Estimating vs Computing**

A healthcare administrator prompts an assistant to analyze a patient readmission spreadsheet. Suspecting that the model produced a glance-based estimate because the response arrived in under two seconds with no code block, which corrective prompt should she issue?

A) 'Please re-read the file with higher attention focus and tell me if you feel confident in your medical intuition.'
B) 'Are you running code on this file or estimating? If estimating, stop immediately and run code instead.'
C) 'Clear your context memory cache and explain the historical origins of hospital readmission penalty statutes.'
D) 'Translate the readmission spreadsheet into an HTML table so that the browser can compute the column totals.'

QUESTION 12 **Domain 2: Prompt Engineering & In-Context Learning • The Five-Section Brief: Structure & Absence of Technical Jargon**

A non-technical warehouse supervisor writes a brief for an inventory reordering script. A programmer colleague advises the supervisor to rewrite the prompt to specify exact Python dictionary keys, nested for-loops, and memory pointers. Why is specifying the desired output more effective than prescribing programming constructs?

A) Including programming terminology causes cloud inference servers to charge higher enterprise token billing rates.
B) Foundation models treat technical programming keywords as adversarial attack vectors, triggering prompt filters.
C) Programming languages prohibit the use of English words when defining variable allocations in automated scripts.
D) Over-managing implementation how degrades briefs, while focusing purely on what done looks like yields better code.

QUESTION 13 **Domain 2: Prompt Engineering & In-Context Learning • Rules Section: Where Professional Domain Knowledge Lives**

A retail operations coordinator briefs an AI script to analyze customer returns across five regional branches. The script incorrectly flags dozens of valid returns as fraudulent. What omission in the five-part brief caused this failure?

A) Omitting the Output part requesting that the final fraud analysis report be styled in dark green typography.
B) Omitting explicit instructions requiring the AI to write the sorting algorithms in low-level C programming code.
C) Omitting the Rules part containing un-inferable company policies, such as a 60-day holiday return extension.
D) Omitting the Input part specifying the byte size and hard disk sector address of the returns spreadsheet.

QUESTION 14 **Domain 2: Prompt Engineering & In-Context Learning • Edge Cases Section: Proactive Handling of Imperfect Data**

A clinical laboratory technician scripts an automated blood sample analysis rollup. When the script encounters a corrupted record containing an alphanumeric character in the numerical white blood cell field, it silently converts the field to zero, distorting the clinical averages. How should the Edge Cases block have been structured?

A) Require the script to crash immediately and delete the original laboratory CSV file to protect patient safety.
B) Command the script to extrapolate missing medical values by searching public online medical encyclopedias.
C) Mandate that any unparseable or corrupted row must be skipped without guessing, and listed in a final report.
D) Instruct the script to replace corrupted medical values with the global median of the entire patient population.

QUESTION 15 **Domain 2: Prompt Engineering & In-Context Learning • Pre-Brief Discovery: Inspect Before Briefing**

A university admissions officer receives a messy legacy CSV export of 5,000 international student applications. Before writing a five-part code brief to calculate scholarship eligibility, the officer is unsure what data formatting anomalies exist. What exploratory move should the officer execute before writing the final script brief?

A) Manually open all 5,000 application rows in a text editor to inspect every individual column cell by hand.
B) Write a script that automatically deletes any application containing non-standard foreign address characters.
C) Prompt the AI to examine the file first and report any formatting inconsistencies or unexpected column values.
D) Instruct the AI to guess reasonable scholarship eligibility criteria based on standard domestic guidelines.

QUESTION 16 **Domain 4: Model Evaluation, Validation & Error Mitigation • The Known-Answer Test: The Non-Negotiable Firewall**

An accountant commissions an AI-written script to reconcile 15,000 banking transactions spanning an entire fiscal year. Before trusting the script's annual report, she runs the script on the month of March, which she previously reconciled by hand and knows totaled exactly $184,250. What verification technique is she employing?

A) The known-answer test, establishing an empirical firewall by testing the script against an independently verified benchmark.
B) A reality check, counting whether the total number of banking transactions exceeds the annual debit card quota.
C) An adversarial pass, demanding that the model identify hypothetical vulnerabilities in federal banking statutes.
D) A cross-model check, comparing how two competing neural network architectures tokenize financial accounting ledgers.

QUESTION 17 **Domain 4: Model Evaluation, Validation & Error Mitigation • Injecting Known Faults: Mutation Testing in Practice**

A hospital pharmacist commissions an AI script to detect discrepancies between dispensed narcotics logs and physical shelf inventory counts. Before running the script on live hospital inventory, she intentionally introduces a deliberate 10-unit deficit into a test copy of the records. What principle does this test embody?

A) Context rot, filling the LLM context window with synthetic errors to test token memory compacting routines.
B) Adversarial prompt injection, attempting to bypass hospital electronic narcotics dispensing security protocols.
C) Mutation testing, confirming that the verification script actually detects a known fault before trusting it on real data.
D) Data poisoning, corrupting base foundation model weights to prevent unauthorized surveillance of inventory logs.

QUESTION 18 **Domain 4: Model Evaluation, Validation & Error Mitigation • Reality Questions: Checking Invariants and Row Counts**

A human resources analyst reviews an AI-generated payroll summary report. The input spreadsheet contained 1,400 employee records, but the final departmental breakdown table only accounts for 1,381 employees. Which verification method caught this discrepancy, and what should the analyst do next?

A) Reality questions; ask the AI to explain where the 19 missing rows went and show the filtering logic applied.
B) Adversarial pass; prompt the model to adopt a hostile defense attorney persona to cross-examine HR policies.
C) Cross-model check; paste the 19 missing employee names into a public search engine to find their social media accounts.
D) Known-answer test; re-calculate all 1,400 employee payroll tax withholdings by hand using paper ledger tables.

QUESTION 19 **Domain 4: Model Evaluation, Validation & Error Mitigation • Plain-English Replay: Auditing Algorithmic Procedures**

A school administrator cannot read Python code, but must ensure that an AI-written script accurately identifies students eligible for honors diplomas. The administrator instructs the AI: 'Explain step-by-step in plain English what your code does, including how you handle incomplete semesters.' The AI explains that it averaged incomplete semesters as 0% grades. Why is this verification method so effective?

A) Demanding a plain-English explanation bypasses the model's token limits, allowing infinite context ingestion.
B) Most real errors are procedural misunderstandings that become immediately obvious when explained in plain English.
C) Plain-English replays legally transfer all institutional diploma certification liability directly to the AI provider.
D) Plain-English replays automatically recompile the underlying Python script into verified binary C++ machine code.

QUESTION 20 Domain 4: Model Evaluation, Validation & Error Mitigation • Uncle Bob Martin & Extreme Constraints

A senior engineering director cites Robert C. Martin ('Uncle Bob') to justify why non-programmers can trust AI-written code without reading syntax. What crucial nuance regarding Uncle Bob's transition must the director emphasize?

A) He concluded that modern foundation models are mathematically incapable of generating software bugs or security flaws.
B) He determined that reading code was an outdated historical artifact that should be formally banned in software firms.
C) He stopped reading code only because he surrounded agents with extreme constraints and automatic verification checks.
D) He proved that natural language prompts automatically eliminate the need for software testing or quality assurance.

QUESTION 21 Domain 4: Model Evaluation, Validation & Error Mitigation • Red-Text Errors: Errors as Precise Dialogue

A marketing specialist prompts an AI assistant to execute a customer segmentation script. The execution halts abruptly, displaying an intimidating 15-line red Python traceback mentioning `KeyError: 'Postal_Code'`. How should the specialist respond?

A) Re-type the error message manually into a public search engine to find software patch downloads for Windows.
B) Forward the entire red error traceback back to the AI without interpreting it, commanding it to diagnose and fix.
C) Immediately delete the customer database and conclude that AI is incapable of processing marketing records.
D) Enroll in an online computer science program to learn how Python dictionary key-value data structures function.

QUESTION 22 Domain 4: Model Evaluation, Validation & Error Mitigation • The 'Wrong-but-Running' Silent Failure

A sales manager runs an AI script on an annual sales ledger. The script displays a green 'Executed Successfully' banner with zero errors, reporting total annual revenue of $4.2 million. However, the manager knows from quarterly board meetings that revenue was approximately $12 million. What diagnostic prompt should the manager issue?

A) Type 'This is completely wrong, fix it immediately' without providing any context, symptoms, or expected values.
B) Manually add $7.8 million to the script's final total to force the revenue figure to match the quarterly reports.
C) Clear all chat history and assume the company experienced a catastrophic, unrecorded 65% revenue collapse.
D) Report the symptom and expected value, and instruct the AI to show row counts, date ranges, and sample rows seen.

QUESTION 23 — Domain 4: Model Evaluation, Validation & Error Mitigation • The Three-Strikes Escalation Rule

A logistics dispatcher attempts to script a delivery route optimizer. For three consecutive turns, the script crashes on the same date parsing error despite the assistant proposing minor code tweaks each time. What operational escalation rule should the dispatcher execute?

A) Upgrade the computer's physical motherboard RAM memory modules to expand the operating system date parsing buffer.
B) Step back and instruct the AI to restate the problem fresh and propose two completely different logical approaches.
C) Manually delete all date timestamps from the delivery database to force the script to run without temporal logic.
D) Continue pasting the identical error message repeatedly until the model's neural network randomly discovers a fix.

QUESTION 24 — Domain 4: Model Evaluation, Validation & Error Mitigation • Unfamiliar Terminology in Errors (The Codec Principle)

A corporate communications coordinator runs a text-cleaning script. The execution crashes with an error: `UnicodeDecodeError: 'utf-8' codec can't decode byte 0x96`. The coordinator has never heard of a codec or byte encoding. What is the most effective operational move for the coordinator?

A) Rename the file extension from `.csv` to `.exe` to bypass operating system character decoding verification checks.
B) Abandon the project because encountering encoding errors indicates that the underlying text files are irreparably damaged.
C) Spend the afternoon researching character encoding standards from the 1980s before attempting to re-run the script.
D) Paste the entire error into the chat without worrying about definitions; the AI understands technical error vocabulary.

QUESTION 25 — Domain 6: Enterprise Knowledge Systems & RAG • The Three-Part Script Retention Habit

An accounts payable clerk successfully briefs an AI assistant to identify duplicate vendor invoices. To ensure that this solved problem becomes a permanent, reusable operational button rather than a lost one-time chat session, what three-part retention habit should the clerk implement?

A) Take a screenshot of the chat window, print it on paper, and file it in an offsite physical records storage vault.
B) Save the script file, keep the plain-English brief alongside it, and store them with a small sample test file.
C) Copy the raw Python code into an email and send it to all departmental staff with instructions to learn coding.
D) Memorize the prompt wording so that it can be retyped manually into a fresh chat window at the end of every month.

QUESTION 26 Domain 6: Enterprise Knowledge Systems & RAG • The Intent Layer: Pairing Brief.md with Script.py

Six months after building an automated student grade calculation script, a university department alters its late-submission penalty policy. The department administrator opens the `grade-calculator/` folder containing `brief.md` and `calculate.py`. How does the presence of `brief.md` (the Intent Layer) facilitate updating the tool?

A) The administrator uses `brief.md` to manually edit the Python variable assignment statements line by line in notepad.
B) The administrator hands both files to an AI, states the rule change in English, and lets the AI update the code.
C) The presence of `brief.md` automatically disables the script's execution until an external software engineer signs off.
D) The administrator deletes `calculate.py` and completely rebuilds the grading system from scratch in an unorganized chat.

QUESTION 27 Domain 6: Enterprise Knowledge Systems & RAG • Preventing Rule Drift Across Time

A medical clinic's no-show appointment analysis is executed by different administrative assistants each month. In May, Assistant A prompts the AI conversationally; in June, Assistant B prompts it using slightly different wording, resulting in divergent no-show rates. How does saving and reusing a script prevent this operational failure?

A) The exact business rules and edge-case definitions are frozen in the script, ensuring identical analytical logic every run.
B) Saved scripts automatically force hospital patients to arrive thirty minutes early for all scheduled clinic visits.
C) Saved scripts prohibit administrative assistants from accessing patient files unless supervised by a physician.
D) Saved scripts permanently lock the clinic's computer operating systems to prevent installing external software.

QUESTION 28 Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • Rerunning Scripts Across Environments

A research assistant saves a Python data cleanup script generated inside a web browser chat. The following month, the assistant has a new batch of 50 data files located in a local desktop folder. What core architectural boundary exists between browser chat sandboxes and local workstations?

A) Scripts authored in browser sandboxes are mathematically encrypted and cannot execute on local desktop computers.
B) Local desktop computers are incapable of running Python scripts unless connected to a continuous cloud chat session.
C) Browser chat sandboxes automatically synchronize all local desktop folders every twenty-four hours in the background.
D) Chat sandboxes are temporary workshops where scripts are born; the local machine is where persistent scripts live.

QUESTION 29 **Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • Browser Sandboxes: Zero-Risk Learning Workshop**

A financial auditor on a borrowed corporate laptop needs to run a quick reconciliation on three quarterly expense spreadsheets. The auditor cannot install software or access command-line terminals. Why is the browser chat sandbox (Claude.ai, ChatGPT, Gemini) the ideal execution surface?

A) It executes code in a secure cloud sandbox on the provider's servers, posing zero risk to the local computer.
B) It permanently registers the auditor's identity with international accounting standards regulatory boards.
C) It allows the cloud model to access and reconfigure the local laptop's operating system registry keys.
D) It automatically bypasses all corporate data classification policies by encrypting file names during upload.

QUESTION 30 **Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • Terminal Agents: The 'Chat Box in a Folder' Mental Model**

A non-programming clinical researcher hears that tools like Claude Code and OpenCode operate inside a computer terminal and assumes they are exclusively for software engineers. How is a local terminal agent properly understood in contrast to a browser chat session?

A) The terminal automatically translates spoken human voice commands into low-level electrical transistor pulses.
B) The terminal functions as a plain-English chat box situated directly inside a folder, eliminating manual uploads.
C) The terminal is an automated software bot that replaces the researcher by signing clinical trial publications.
D) The terminal forces the user to write binary machine bytecode before granting access to local spreadsheet files.

QUESTION 31 **Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • Desktop Apps: Plan-Then-Approve Workflow**

A corporate archivist uses an AI desktop application (Cowork or OpenWork) to reorganize 500 scanned historical contracts. Before moving or renaming any files, the application pauses and presents a complete structured plain-English list of proposed actions. What defining architectural workflow does this represent?

A) Plan-then-approve, moving procedural plain-English replay before execution so assumptions are verified prior to action.
B) Continuous integration build verification, compiling the archive directory into a standalone software application.
C) Automated code obfuscation, hiding underlying script logic to prevent the archivist from inspecting file renames.
D) Hardware-level disk defragmentation, reallocating magnetic sectors before executing file moving subroutines.

QUESTION 32 **Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • The Graduation Signal: When Uploads Annoy You**

A marketing analyst has spent three months using browser chat sandboxes to generate weekly ad spend reports. She now finds that manually uploading forty regional campaign CSV files every Monday morning is tedious and error-prone. What operational milestone does this workflow bottleneck represent?

A) She must purchase a private cloud mainframe server to bypass browser file upload network transfer limitations.
B) She must instruct the AI assistant to automatically merge her personal credit card with corporate ad billing accounts.
C) She has reached the graduation signal to transition from browser sandboxes to terminal or desktop local folder agents.
D) She has reached the technical ceiling of generative artificial intelligence and must return to manual Excel formulas.

QUESTION 33 **Domain 5: AI Ethics, Governance & Diligence • Blast Radius: Definition & Irreversible Deletion Risks**

A junior IT technician permits an autonomous desktop agent to 'clean up obsolete files' across an entire departmental file server. In under thirty seconds, the agent deletes 400 customer order records. The technician immediately checks the Windows Recycle Bin, but finds it completely empty. Why did this occur?

A) The foundation model detected proprietary customer data and initiated an automated anti-piracy self-destruct sequence.
B) Programmatic file deletions executed by scripts frequently bypass the operating system's Recycle Bin entirely.
C) The operating system automatically encrypts files deleted by artificial intelligence in hidden cloud quarantine vaults.
D) The file server experienced an unexpected electrical power surge that demagnetized the physical storage media.

QUESTION 34 **Domain 5: AI Ethics, Governance & Diligence • Rule 1 & Rule 4: Copies First & Writing to New Files**

A laboratory data coordinator briefs a script to standardize raw spectrometer readings: 'Standardize temperature values, convert Fahrenheit to Celsius, and save changes.' What critical safety directives must the coordinator mandate to minimize blast radius?

A) Copy the original folder to a backup first, write all processed results to new files, and leave originals untouched.
B) Require the script to delete all spectrometer readings older than seven days to optimize data processing speeds.
C) Instruct the script to overwrite the master raw data files in place to save local solid-state drive storage space.
D) Disable operating system file permissions to allow the AI script to execute with unrestricted administrator rights.

QUESTION 35 **Domain 5: AI Ethics, Governance & Diligence • Rule 2: Demanding Dry Runs Before File Operations**

An academic researcher instructs an AI agent to clean up a directory of audio interview recordings by deleting duplicate files. In accordance with Rule 2, the researcher demands a dry run first. The dry-run plan lists 40 proposed deletions, but inspection reveals six files are unique interviews that happened to have similar file sizes. How did the dry run save the project?

A) It permanently revoked the AI agent's access to the computer's hard drive to prevent any future script runs.
B) It displayed the complete list of proposed file operations, allowing the researcher to catch flawed rules before action.
C) It proved that the AI agent possessed native human auditory perception capable of transcribing speech without text.
D) It automatically created compressed audio duplicates of the six unique interviews on a secure external server.

QUESTION 36 **Domain 5: AI Ethics, Governance & Diligence • Rule 3: Scoping to the Smallest Folder Boundary**

A legal clerk opens a terminal agent to organize contracts for a specific client named Acme Corp. Instead of opening the terminal inside `Documents/Clients/AcmeCorp/`, the clerk opens the terminal in the root `C:\Users\Clerk\` home directory and commands: 'Reorganize all contract files by date.' What severe risk did the clerk introduce?

A) A confused script running at the user root directory can traverse, rename, or damage personal files across the entire drive.
B) The foundation model will immediately transmit the user's personal tax returns to public internet search engines.
C) The terminal agent will automatically exceed Windows RAM capacity and trigger a physical motherboard hardware short.
D) The operating system will automatically flag the legal clerk's computer as a rogue cryptocurrency mining node.

QUESTION 37 **Domain 7: Specialized Agent Architectures & SDD • The Edge of the Map: Where One-Prompt Code Ends**

A startup founder attempts to use a single prompt in a chat assistant to build an entire multi-tenant commercial SaaS banking application featuring user authentication, payment processing, regulatory compliance auditing, and 99.99% high availability. Why is this project classified as beyond the scope of a single-prompt script?

A) Foundation models are legally prohibited by international banking treaties from generating web software applications.
B) Programming languages like Python cannot execute mathematical financial accounting transactions in cloud browsers.
C) Browser chat sandboxes automatically terminate any user session that attempts to write more than 50 lines of code.
D) Multi-user authentication, live payments, and unattended automation are production software systems, not simple scripts.

QUESTION 38 **Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • Judgment Wearing a Computation Costume**

A corporate HR committee asks an AI assistant to: 'Write and run Python code to analyze employee productivity metrics and calculate exactly which three staff members should be terminated this quarter.' Why is this request an inappropriate delegation of authority?

A) As an abdication of human responsibility; the math is trivial, but the ethical and strategic criteria belong to humans.
B) As an exemplary application of data-driven computational objectivity that eliminates human cognitive bias in firing.
C) As an ideal code problem that should be delegated entirely to an autonomous Digital FTE operating without oversight.
D) As an impossible technical calculation because employee productivity metrics cannot be represented as numbers.

QUESTION 39 **Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • The Off-the-Shelf App Test: A Market of One**

A university student wants to calculate his semester grade. His academic grading guide has a bespoke rule: 'Homework is 20% (drop 2 lowest), quizzes are 20% (best 8 of 10 count), midterm is 25%, and final is 35% (if final is higher than midterm, final replaces midterm).' Commercial gradebook apps cannot handle this calculation. Why is this problem uniquely suited for a custom-generated script?

A) The student can force the AI model to hack the university's central database and alter official semester transcripts.
B) Commercial gradebook applications are legally prohibited from calculating weighted grade averages in universities.
C) Writing a custom Python script automatically registers the student's grading formula with educational app stores.
D) It represents a bespoke calculation where rules exist only in these guidelines, making plain-English briefs uniquely powerful.

QUESTION 40 **Domain 3: AI Augmentation, Task Delegation & Agentic Workflows • The Project Catch: Actionable Truth with a Real Person**

In Project 3 ('The Books Don't Match'), an organizer reconciles 4,000 in trip dues against a messy mobile payment log. The code reports that 3,500 was collected, identifying that seven people paid, one mystery 500 payment had no name, and Usman only paid 300 with a memo reading 'pizza'. What represents the true final deliverable of this project?

A) An actionable sentence spoken to a real person: messaging Usman to verify if the mystery 500 payment was his trip dues.
B) Publishing the raw 40-line Python reconciliation script to a public open-source Git code repository for review.
C) Filing a formal police fraud complaint against the mobile payment service provider for misplacing trip dues.
D) Formatting the bank transaction logs into an elaborate multi-page marketing brochure to distribute to classmates.

PART 2: ANSWER KEY & COMPREHENSIVE RATIONALES

Exhaustive psychometric explanations detailing why the correct option is uniquely accurate per the chapter source text, and why each alternate distractor is operationally flawed.

Question 1: Correct Option B

Target Domain: Domain 1: AI Foundations & Architecture | **Grounding:** Concept 1: Code is no longer gated by coding (Lines 132-150)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 1. The chapter explains: 'The new deal: AI is the developer, you are the client, and the developer works in seconds, for free, and never gets tired of your revisions... You are not learning to write code. You are learning to be a good client for code. Good clients do not lay bricks. They write clear briefs, check the work against what they can measure, and keep what they paid for.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because real-world clinical databases frequently contain disparate entries and scheduling conflicts that require active reconciliation.
- **Option C Flaw:** Option C is incorrect because Python remains the primary language for data and file processing; it is generated by AI rather than typed manually by the client.
- **Option D Flaw:** Option D is incorrect because programming bootcamps face no legal prohibitions against teaching data reconciliation.

Question 2: Correct Option A

Target Domain: Domain 1: AI Foundations & Architecture | **Grounding:** Concept 2: Code is exact, in both directions (Lines 232-236)

Source Grounding & Rationale: Option A is uniquely accurate per Concept 2. The text highlights this exact double-edged property: 'Code is exact, in both directions. It computes your totals to the last cent with no arithmetic errors. It also does exactly what it was told, including the wrong thing, perfectly, at scale. A script built on a misunderstanding renames 300 files wrongly just as fast as it renames them rightly. Power and danger come from the same property.'

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because the failure stems from logical specification error, not physical neural weight memory corruption.
- **Option C Flaw:** Option C is incorrect because operating systems execute batch file operations reliably; the disaster was caused by the logic of the script.
- **Option D Flaw:** Option D is incorrect because Python scripts are interpreted directly in execution environments without manual conversion to assembly code.

Question 3: Correct Option A

Target Domain: Domain 1: AI Foundations & Architecture | **Grounding:** Concept 2: Code comes in many languages (Lines 211-220)

Source Grounding & Rationale: Option A is uniquely accurate per Concept 2. The chapter emphasizes: 'You never choose between them. Picking the language is a technical judgment, and technical judgments are what you are handing over. A number from a spreadsheet means Python. A button someone can click means JavaScript. You only need the two names so you recognize what is scrolling past.'

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because demanding C++ or SQL micromanages technical execution and complicates simple browser-based tools.
- **Option C Flaw:** Option C is incorrect because browsers natively execute JavaScript/TypeScript for clickable interfaces, not raw Python scripts.
- **Option D Flaw:** Option D is incorrect because student privacy is governed by data handling and sandbox boundaries, not by requiring Rust code.

Question 4: Correct Option C

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Concept 3: The dividing line: which problems are code problems (Lines 243-285)

Source Grounding & Rationale: Option C is uniquely accurate per Concept 3. The chapter introduces the VPRF framework: Volume (more items than you'd do by hand), Precision (a wrong digit has consequences), Repetition (facing it again), and Files (lives in files, not sentences). Any single signal marks a code problem. Totalling 4,500 rows across 12 spreadsheets fires Volume, Precision, and Files. The others are answer problems suited for conversational reasoning.

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because drafting an email is conversational mind-work (an answer problem), not hands-work on tabular files.
- **Option B Flaw:** Option B is incorrect because judging a brand slogan is subjective qualitative reasoning, not deterministic programmatic computation.
- **Option D Flaw:** Option D is incorrect because summarizing a single short memo is an answer problem that models handle directly in context without code.

Question 5: Correct Option D

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Concept 3: The trap to watch for & School administrator case study (Lines 341-368)

Source Grounding & Rationale: Option D is uniquely accurate per Concept 3. The text warns: 'The trap to watch for: problems that look like answer problems but fire Precision. "Roughly how did my sales trend?" feels conversational, so people ask conversationally and get an approximate answer... The AI had answered from a skim, and a skim reads a partial payment as paid. One sentence would have prevented it: "write and run code to find them."'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because the failure was caused by skipping code execution, not missing pre-trained domain facts.
- **Option B Flaw:** Option B is incorrect because cloud model inference accuracy does not fluctuate based on the day of the week.
- **Option C Flaw:** Option C is incorrect because converting tabular spreadsheets into vector embeddings degrades numerical precision rather than improving it.

Question 6: Correct Option A

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Concept 3: Repetition signal & Concept 8 (Lines 271-276, 335-340, 610-613)

Source Grounding & Rationale: Option A is uniquely accurate per Concept 3 and Concept 8. The text highlights: 'Repetition, the strongest signal. The script becomes an asset... A script written once is something you own forever. The teacher's submission-check brief takes effort the first time. On the next assignment the same job is one sentence: "Run my script on this new file." This is the moment you stop using AI and start building up with it.'

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because code generation does not trigger automatic corporate copyright filings.
- **Option C Flaw:** Option C is incorrect because executing Python scripts does not modify the frozen weights of foundation models.
- **Option D Flaw:** Option D is incorrect because running automated scripts does not alter commercial API billing terms or waive token charges.

Question 7: Correct Option B

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Concept 4: The opposite matters too (Lines 436-438)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 4. The chapter cautions: 'The opposite matters too. Do not demand code for answer problems. "Write and run code to tell me if my essay is persuasive" wastes the tool. Concept 3's dividing line works in both directions.' Empathy and cultural fit are subjective qualitative evaluations (mind-work), not mechanical arithmetic (hands-work).

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because criminal background checks are legally and ethically inappropriate for assessing cover letter empathy.
- **Option C Flaw:** Option C is incorrect because changing programming languages does not convert a qualitative judgment into a deterministic arithmetic task.
- **Option D Flaw:** Option D is incorrect because Kubernetes infrastructure is completely irrelevant to qualitative prompt design.

Question 8: Correct Option D

Target Domain: Domain 2: Prompt Engineering & In-Context Learning | **Grounding:** Concept 4: Make the AI's hands move & The three lines (Lines 390-403)

Source Grounding & Rationale: Option D is uniquely accurate per Concept 4. The chapter provides the non-negotiable three-line standard: 'Write and run code to answer this. Show me the code you ran. Before analyzing anything, tell me the exact row count, the column names, and the date range of the file.' Line 1 forces code, Line 2 provides visual proof, and Line 3 acts as the cheapest lie detector to verify that the file was actually opened and parsed.

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because extrapolating annual totals from ten rows is the definition of ungrounded estimation.
- **Option B Flaw:** Option B is incorrect because temperature zero does not force code execution, and translating numbers to Latin adds meaningless noise.
- **Option C Flaw:** Option C is incorrect because role-prompting an accountant persona does not force code execution; the model can still hallucinate estimates authoritatively.

Question 9: Correct Option B

Target Domain: Domain 4: Model Evaluation, Validation & Error Mitigation | **Grounding:** Concept 4: The tell (Lines 438-441)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 4 ('The tell'). The chapter states: 'A reply about your data with no code block, no row counts, and no "here is what I computed" is an opinion about your data, not an analysis of it. Treat it the way you would treat a builder's invoice for work you never saw.' Without a code block, the model merely predicted plausible numbers.

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because sandboxed execution environments explicitly display code execution badges or code blocks when tools fire.
- **Option C Flaw:** Option C is incorrect because computing overtime is a standard spreadsheet automation task with no inherent legal prohibition.
- **Option D Flaw:** Option D is incorrect because LLM chat responses are plain text; they do not conceal calculations inside encrypted packets.

Question 10: Correct Option C

Target Domain: Domain 2: Prompt Engineering & In-Context Learning | **Grounding:** Concept 4: Say 'code', not 'Python' (Lines 401-404)

Source Grounding & Rationale: Option C is uniquely accurate per Concept 4. The text explains: 'Notice those lines say "code", not "Python". That is the rule for every prompt in this course. You tell the AI to compute. You never tell it which language to compute in. Adding "in Python" forces its hand on a guess you are not equipped to make. On the rare problem where Python is the wrong choice, you would have steered it wrong without knowing.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because foundation models are multilingual across code and understand Julia, R, and Python easily.
- **Option B Flaw:** Option B is incorrect because sandboxes support multiple runtimes or will politely explain environment limits without session termination.
- **Option D Flaw:** Option D is incorrect because standard open-source programming languages (Python, Julia, R) have no licensing fees.

Question 11: Correct Option B

Target Domain: Domain 2: Prompt Engineering & In-Context Learning | **Grounding:** Concept 4: Phrasing comparison table (Lines 405-435)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 4 table. The text lists this exact diagnostic move: '"Are you running code on this file, or estimating? If estimating, stop and run code instead." The AI names its method, then computes. The best move when you suspect a glance.' It forces the model to acknowledge its methodology and execute programmatic analysis.

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because asking about 'confidence' or 'medical intuition' invites sycophantic praise without forcing code execution.
- **Option C Flaw:** Option C is incorrect because clearing cache to discuss historical statutes shifts the task to an irrelevant trivia query.
- **Option D Flaw:** Option D is incorrect because generating an HTML table displays text; it does not calculate statistical readmission sums.

Question 12: Correct Option D

Target Domain: Domain 2: Prompt Engineering & In-Context Learning | **Grounding:** Concept 5: Brief the problem, not the program (Lines 452-465)

Source Grounding & Rationale: Option D is uniquely accurate per Concept 5. The chapter states: 'The best code briefs contain no technical language at all. You do not name loops, libraries, file handling, or the programming language. You describe what a good outcome looks like, exactly as you would for a human assistant. People who half-know programming often write worse briefs than total beginners, because they over-manage the how and under-describe the what.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because token pricing is uniform regardless of whether words are technical or conversational.
- **Option B Flaw:** Option B is incorrect because technical keywords are not classified as adversarial attack vectors.
- **Option C Flaw:** Option C is incorrect because modern programming languages are built entirely on English keywords.

Question 13: Correct Option C

Target Domain: Domain 2: Prompt Engineering & In-Context Learning | **Grounding:** Concept 5: Rules is where your professional knowledge lives (Lines 466-469)

Source Grounding & Rationale: Option C is uniquely accurate per Concept 5. The text emphasizes: 'Rules is where your professional knowledge lives. The AI knows Python. It does not know that "Pending" means not yet submitted in your system, or that the downtown store closed in March. Every wrong analysis you will ever get traces back to a rule you knew and did not state.' Without stating the 60-day holiday extension, the model applied standard generic return rules.

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because font typography styling does not affect the logical calculation of return fraud.
- **Option B Flaw:** Option B is incorrect because the programming language chosen by the AI is irrelevant to missing business return rules.
- **Option D Flaw:** Option D is incorrect because hard disk sector addresses are low-level operating system details that have zero bearing on return policies.

Question 14: Correct Option C

Target Domain: Domain 2: Prompt Engineering & In-Context Learning | **Grounding:** Concept 5: Edge cases: skip and list (Lines 468-470)

Source Grounding & Rationale: Option C is uniquely accurate per Concept 5. The chapter specifies the professional standard for Edge cases: 'Edge cases is where you decide, in advance, what imperfect data should do. Your data is imperfect. Everyone's is. The professional move is one sentence: "If you hit a row you cannot read, do not guess. Skip it, and list every skipped row at the end." That turns silent damage into a visible report.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because deleting original laboratory data causes irreversible data loss.
- **Option B Flaw:** Option B is incorrect because searching online encyclopedias cannot recover patient-specific empirical blood counts.
- **Option D Flaw:** Option D is incorrect because imputing median values into clinical blood panels without physician awareness creates dangerous medical distortions.

Question 15: Correct Option C

Target Domain: Domain 2: Prompt Engineering & In-Context Learning | **Grounding:** Concept 5: Don't know your own edge cases? Ask. (Lines 478-481)

Source Grounding & Rationale: Option C is uniquely accurate per Concept 5. The text advises: 'Don't know your own edge cases? Ask. You often will not know what is hiding in your data. So make discovery the first prompt: "Before doing anything, examine the files and tell me what could be unclear, inconsistent, or surprising in here." The AI comes back with... Now you write the Edge cases section with knowledge instead of imagination. Inspect, ask, then brief.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because manually inspecting 5,000 rows defeats the purpose of AI automation and wastes dozens of hours.
- **Option B Flaw:** Option B is incorrect because deleting foreign applicant records is discriminatory and causes catastrophic data destruction.
- **Option D Flaw:** Option D is incorrect because guessing eligibility criteria abdicates administrative policy and introduces bias.

Question 16: Correct Option A

Target Domain: Domain 4: Model Evaluation, Validation & Error Mitigation | **Grounding:** Concept 6: Verifying work you cannot read & Known-answer test (Lines 490-503, 552-555)

Source Grounding & Rationale: Option A is uniquely accurate per Concept 6. The text establishes: 'The single most powerful move on this page is the known-answer test, which means running the code on a small part whose answer you already know. Run the script on one month you have already closed by hand... If the script reproduces every answer you can check, your confidence in the rest is earned, not hoped for... Never act on a precision-critical number you have not tested against one answer you independently know.'

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because checking row counts is a reality question, not a benchmarked known-answer test.
- **Option C Flaw:** Option C is incorrect because an adversarial pass asks the AI to attack its own assumptions, rather than running a known numerical test.
- **Option D Flaw:** Option D is incorrect because a cross-model check compares two different AI models, not a script against a human-calculated benchmark.

Question 17: Correct Option C

Target Domain: Domain 4: Model Evaluation, Validation & Error Mitigation | **Grounding:** Concept 6: Pharmacist case study & Uncle Bob Martin's mutation testing (Lines 534-551)

Source Grounding & Rationale: Option C is uniquely accurate per Concept 6. The chapter details the pharmacist example and connects it to software engineering: 'In mutation testing you break the main code on purpose, one small change at a time, and see whether any test notices. A test that stays silent while the code is wrong has just proved that it proves nothing. Now look at the pharmacist again... Add one known error to your data and see if the code finds it. That is the known-answer test in its most elegant form, and it costs ninety seconds.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because introducing a test record is a deliberate validation technique, not accidental conversational context rot.
- **Option B Flaw:** Option B is incorrect because testing an internal script on a data copy is rigorous quality assurance, not a malicious prompt injection attack.
- **Option D Flaw:** Option D is incorrect because modifying test data in an isolated session does not alter cloud base model weights.

Question 18: Correct Option A

Target Domain: Domain 4: Model Evaluation, Validation & Error Mitigation | **Grounding:** Concept 6: Reality questions (Lines 504-505)

Source Grounding & Rationale: Option A is uniquely accurate per Concept 6. The text explains: 'Reality questions are questions you put to the output using what you already know. Count rows in versus rows out. If 1,400 went in and the report covers 1,381, where are the missing 19? Does the "biggest customer" match the one you know is biggest? Any answer that contradicts your professional sense deserves the next prompt: "Your report says X. Walk me through in plain English how you computed that number, and show me three of the rows behind it."'

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because role-playing a hostile attorney does not diagnose which data rows were filtered out.
- **Option C Flaw:** Option C is incorrect because searching employee names on public search engines is a severe breach of employee privacy.
- **Option D Flaw:** Option D is incorrect because manually calculating 1,400 payroll records by hand is an absurd waste of time; row-count reconciliation is the immediate step.

Question 19: Correct Option B

Target Domain: Domain 4: Model Evaluation, Validation & Error Mitigation | **Grounding:** Concept 6: Plain-English replay (Lines 506-511)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 6. The chapter states: 'Plain-English replay means the AI's own account, in plain words, of what its code did... You cannot check Python, but you can check "I matched on amount... and treated blanks as zero." If the procedure is wrong, through a wrong rule or a wrong assumption, it reads wrong in English too, and you catch it. Most real errors are procedure errors, not typing errors, which is why this works.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because plain-English replays consume standard context tokens and do not bypass window limits.
- **Option C Flaw:** Option C is incorrect because institutional accountability is non-transferable and remains entirely with the school administrator.
- **Option D Flaw:** Option D is incorrect because natural language explanations do not recompile scripts into C++ code.

Question 20: Correct Option C

Target Domain: Domain 4: Model Evaluation, Validation & Error Mitigation | **Grounding:** Concept 6: The man who wrote the book on reading code stopped reading code (Lines 536-551)

Source Grounding & Rationale: Option C is uniquely accurate per Concept 6. The text explains: 'Robert C. Martin... posted that he no longer reads the code his agents write. Instead he surrounds them with what he calls "extreme constraints", which are layers of automatic checks the code has to survive before he believes it... But notice the order, because this is what most people online missed. He did not stop reading and then hope for the best. He built the checks first. Not reading was what the checks allowed.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because LLMs frequently generate bugs, which is precisely why automated checks and constraints are mandatory.
- **Option B Flaw:** Option B is incorrect because Uncle Bob did not ban reading code; he adopted agentic automation underpinned by rigorous test harnesses.
- **Option D Flaw:** Option D is incorrect because automated testing is more critical than ever when code is generated rapidly by AI agents.

Question 21: Correct Option B

Target Domain: Domain 4: Model Evaluation, Validation & Error Mitigation | **Grounding:** Concept 7: When it breaks: errors are dialogue, not failure (Lines 563-573)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 7. The chapter emphasizes: 'An error is not the project failing. It is the computer describing, very precisely, what it needs... The entire skill, for red-text errors: "It stopped and showed this error. Diagnose it, fix the code, and run it again: [paste the red text, all of it]". You do not interpret the error. You forward it. The AI reads its own error messages the way a mechanic reads engine noise, fixes the script, and runs it again.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because searching public forums manually is slow and unnecessary when the AI can read its own traceback instantly.
- **Option C Flaw:** Option C is incorrect because red text is routine debugging dialogue, not an unrecoverable failure warranting data deletion.
- **Option D Flaw:** Option D is incorrect because clients do not need to learn internal Python programming mechanics to get errors repaired.

Question 22: Correct Option D

Target Domain: Domain 4: Model Evaluation, Validation & Error Mitigation | **Grounding:** Concept 7: The harder case is wrong-but-running (Lines 574-580)

Source Grounding & Rationale: Option D is uniquely accurate per Concept 7. The text explains: 'The harder case is wrong-but-running, which means no red text but numbers that look wrong. Do not say "it's wrong, fix it". The AI does not know which part. Report it like a symptom to a doctor: "The output shows total sales of 4.2 million, but I know this year was around 12 million. Something is being dropped or misread. Investigate: show me the row count you processed, the date range you found, and the first 5 rows as the code sees them."'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because typing vague complaints gives the AI no diagnostic baseline to discover what data was skipped or misparsed.
- **Option B Flaw:** Option B is incorrect because arbitrarily falsifying financial figures is fraudulent and evades identifying the data parsing bug.
- **Option C Flaw:** Option C is incorrect because an unexpected number is almost always a data parsing error (such as reading only sheet 1 of an Excel file), not an actual company collapse.

Question 23: Correct Option B

Target Domain: Domain 4: Model Evaluation, Validation & Error Mitigation | **Grounding:** Concept 7: Three patterns cover nearly every breakdown (Lines 594-597)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 7. The breakdown table specifies the rule: 'Same fix fails three times in a row: Stop digging. Say: "We have tried this three times. Step back, restate the problem fresh, and propose two completely different approaches." Three strikes means the approach is wrong, not the typing. If the chat has grown long and confused, start a clean one with the brief and what you learned.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because hardware RAM has nothing to do with algorithmic date parsing logic in software.
- **Option C Flaw:** Option C is incorrect because deleting timestamps destroys essential delivery scheduling data.
- **Option D Flaw:** Option D is incorrect because repeating the exact same failed prompt in a stuck context loop wastes tokens and degrades output quality.

Question 24: Correct Option D

Target Domain: Domain 4: Model Evaluation, Validation & Error Mitigation | **Grounding:** Concept 7: Marketer case study & codec principle (Lines 598-599)

Source Grounding & Rationale: Option D is uniquely accurate per Concept 7. The chapter recounts the marketer example: 'A marketer's script crashed with an error that mentioned a "codec". She pasted it without understanding one word of it. The AI replied that one export was saved in a different text encoding, and it adjusted the code. It ran again and worked. Total downtime was forty seconds. She never learned what a codec is. Reading errors is not required. Willingness to paste them is.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because renaming a CSV to an executable (.exe) will corrupt file associations and trigger security warnings.
- **Option B Flaw:** Option B is incorrect because encoding errors are trivial for the AI to fix once reported.
- **Option C Flaw:** Option C is incorrect because clients do not need to study low-level computer science history to resolve text encoding glitches.

Question 25: Correct Option B

Target Domain: Domain 6: Enterprise Knowledge Systems & RAG | **Grounding:** Concept 8: Keep the script: a solved problem becomes a button (Lines 610-625)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 8. The text outlines the three-part habit: '(1) Ask for the script as a file: Save this as a script file, named clearly, with a plain-English header. (2) Keep the brief next to the script: The script is the how; the brief is the what and why (the Intent Layer). (3) Give the pair a home in one folder: Use my-scripts/ with brief.md, script.py, and a sample input file whose correct answer you know.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because paper screenshots cannot be executed programmatically or modified by future AI sessions.
- **Option C Flaw:** Option C is incorrect because emailing raw code forces colleagues to read syntax, violating the core principle of being a client for code.
- **Option D Flaw:** Option D is incorrect because re-explaining tasks from memory causes rule drift and wastes time every month.

Question 26: Correct Option B

Target Domain: Domain 6: Enterprise Knowledge Systems & RAG | **Grounding:** Concept 8: Keep the brief next to the script (Lines 618-620)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 8. The text explains: 'Keep the brief next to the script. The script is the how, for machines. The brief is the what and why, for humans and for future AI sessions. Six months from now you hand any AI both and say "the late-cutoff rule changed, please update it." The previous course called this the Intent Layer, which means your written intentions kept in files, exact enough for an AI to act on.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because the administrator never needs to manually edit Python syntax line by line.
- **Option C Flaw:** Option C is incorrect because documentation files do not lock or disable script runtime execution.
- **Option D Flaw:** Option D is incorrect because discarding the existing script and starting from scratch ignores the retained asset and wastes previous work.

Question 27: Correct Option A

Target Domain: Domain 6: Enterprise Knowledge Systems & RAG | **Grounding:** Concept 8: Without the keep-the-script habit vs with it table (Lines 628-644)

Source Grounding & Rationale: Option A is uniquely accurate per Concept 8 table. The comparison table contrasts: 'Without the keep-the-script habit: Subtle rule drift: May's logic is not quite June's. With it: The rules are frozen in the script, so every run uses identical logic... Hand the folder to a colleague: brief, script, sample. They're productive in minutes.' Freezing rules in code eliminates conversational ambiguity.

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because data analysis scripts do not control physical patient arrival times.
- **Option C Flaw:** Option C is incorrect because data automation does not alter clinic administrative staffing permissions.
- **Option D Flaw:** Option D is incorrect because Python scripts run within standard application environments without locking operating systems.

Question 28: Correct Option D

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Concept 8: Rerunning the script, surface by surface (Lines 626-627)

Source Grounding & Rationale: Option D is uniquely accurate per Concept 8 and Concept 9. The text delivers this definitive architectural aphorism: 'Chat is where scripts are born. Your machine is where they live... The chat sandbox is a workshop you visit. It is not where your files live. Once your problem is about your computer, its folders and its routines, you have outgrown the visit.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because Python scripts downloaded from sandboxes are plain text files that run on any standard Python interpreter.
- **Option B Flaw:** Option B is incorrect because local machines execute Python offline without requiring active chat sessions.
- **Option C Flaw:** Option C is incorrect because cloud chat sandboxes have zero access or background synchronization to local desktop drives.

Question 29: Correct Option A

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Concept 9: Claude.ai: the home surface (and ChatGPT, Gemini) (Lines 664-706)

Source Grounding & Rationale: Option A is uniquely accurate per Concept 9. The text explains: 'When you ask Claude.ai to write and run code, the code runs in a sandbox on Anthropic's side. Your uploads go in. Results, files, and charts come out. Nothing on your computer is touched, which makes this the zero-risk surface to learn on... Zero installation, zero risk to your machine. Perfect for one-off and exploratory jobs.'

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because browser chats do not register user identities with accounting regulatory boards.
- **Option C Flaw:** Option C is incorrect because cloud sandboxes are strictly isolated and cannot touch local operating system registry keys.
- **Option D Flaw:** Option D is incorrect because browser sandboxes do not bypass corporate data governance; sensitive data must follow enterprise policies.

Question 30: Correct Option B

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Concept 10: Claude Code & OpenCode: the agent in your folder (Lines 713-736)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 10. The text clarifies: 'If the word makes you nervous, hold this picture: The terminal is a chat box that sits inside a folder... Once a tool is installed, you type sentences and an agent answers... The agent sees the files already there, with no uploading. It runs code on them, saves scripts that stay forever, and fixes its own errors... Nothing about "chat with an agent that can see this folder" requires knowing code. It requires knowing your files and your problem.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because terminal agents process typed text prompts, not transistor hardware pulses.
- **Option C Flaw:** Option C is incorrect because agents assist with file operations; they cannot replace researchers or sign regulatory documents.
- **Option D Flaw:** Option D is incorrect because users interact with terminal agents using standard plain-English natural language.

Question 31: Correct Option A

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Concept 11: Cowork & OpenWork: delegation with a safety ritual (Lines 737-743)

Source Grounding & Rationale: Option A is uniquely accurate per Concept 11. The chapter highlights: 'Their defining feature is plan-then-approve, which means the app writes a plan and waits for your approval... You read the plan in plain English, catch the wrong assumption before anything runs, and approve. That is Concept 6's plain-English replay, moved before the work instead of after.'

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because organizing archival PDFs is a file-system task, not a software CI/CD compilation build.
- **Option C Flaw:** Option C is incorrect because plan-then-approve is built specifically for human transparency, not obfuscation.
- **Option D Flaw:** Option D is incorrect because file reorganization operates on the file-system layer, not physical magnetic sector defragmentation.

Question 32: Correct Option C

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Concept 11: Choosing among the five & The honest advice on order (Lines 744-784)

Source Grounding & Rationale: Option C is uniquely accurate per Concept 11. The chapter provides clear practical guidance: 'The honest advice on order: Stay on Claude.ai until uploading becomes the annoying part of your week. That annoyance is the graduation signal... Claude Code or OpenCode in the terminal... Cowork or OpenWork as desktop apps... These keep scripts forever and eliminate uploading because the files are already there.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because terminal and desktop agents run on standard laptops without requiring mainframe servers.
- **Option B Flaw:** Option B is incorrect because merging personal credit cards is an absurd violation of corporate financial controls.
- **Option D Flaw:** Option D is incorrect because moving to local folder agents expands automation capabilities rather than returning to manual Excel entry.

Question 33: Correct Option B

Target Domain: Domain 5: AI Ethics, Governance & Diligence | **Grounding:** Concept 12: Blast radius: the rules for code that touches your files (Lines 791-797)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 12. The text explicitly warns: 'On Claude.ai, mistakes are free. The moment code runs on your machine... Code does exactly what it was told, perfectly, at scale, including the wrong thing. Two facts from the prompting course are worth repeating. Files deleted by an agent often skip the recycle bin, and files edited by a script keep no undo history.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because foundation models contain no anti-piracy file self-destruct mechanisms.
- **Option C Flaw:** Option C is incorrect because operating systems do not route script-deleted files to secret quarantine vaults.
- **Option D Flaw:** Option D is incorrect because programmatic deletion is standard software execution, not a physical electrical surge.

Question 34: Correct Option A

Target Domain: Domain 5: AI Ethics, Governance & Diligence | **Grounding:** Concept 12: Rule 1 & Rule 4 (Lines 799-806)

Source Grounding & Rationale: Option A is uniquely accurate per Concept 12. The chapter mandates: 'Rule 1: Work on copies until a script has earned trust: First copy the folder to backup, then work only on the original. One sentence turns the first bad run into a shrug... Rule 4: Write output to new files, never over the originals: Say "Write results to a new file and leave the originals untouched." The original data stays clean, and every run can be undone by deleting the new file.'

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because deleting older data causes unrecoverable research data loss.
- **Option C Flaw:** Option C is incorrect because overwriting original scientific data in place destroys raw experimental baselines permanently.
- **Option D Flaw:** Option D is incorrect because granting unrestricted administrator rights dangerously expands the blast radius.

Question 35: Correct Option B

Target Domain: Domain 5: AI Ethics, Governance & Diligence | **Grounding:** Concept 12: Rule 2: See changes before they happen & Dry runs (Lines 801-802, 814-822)

Source Grounding & Rationale: Option B is uniquely accurate per Concept 12. The text explains: 'Rule 2: See changes before they happen. A dry run is a list of every change the code would make, shown before it makes any. Before renaming, moving, deleting, or overwriting, say "Don't change anything yet. Show me the full list of operations you would perform... and wait for my approval." Reading that list takes two minutes, and it catches the misunderstanding while it is still words instead of actions.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because catching a rule flaw in a dry run allows the client to correct the rule, not permanently ban automation.
- **Option C Flaw:** Option C is incorrect because dry runs check file-system metadata logic, not human auditory consciousness.
- **Option D Flaw:** Option D is incorrect because a dry run displays proposed actions in text; it does not create compressed audio copies.

Question 36: Correct Option A

Target Domain: Domain 5: AI Ethics, Governance & Diligence | **Grounding:** Concept 12: Rule 3: Point the agent at the smallest folder (Lines 803-804)

Source Grounding & Rationale: Option A is uniquely accurate per Concept 12. The chapter warns: 'Rule 3: Point the agent at the smallest folder. Use the smallest folder that contains the problem. Never your home directory, which is the top folder that holds all of your personal files. Never "Documents", and never the whole disk. A script confused inside expenses-2025/ can damage twelve CSVs you backed up... A script confused at the top of your drive can damage your life.'

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because local terminal agents do not broadcast private files to public search engines.
- **Option C Flaw:** Option C is incorrect because running file-listing scripts at root does not cause electrical motherboard short-circuits.
- **Option D Flaw:** Option D is incorrect because directory traversals are standard operating system functions, not cryptocurrency mining.

Question 37: Correct Option D

Target Domain: Domain 7: Specialized Agent Architectures & SDD | **Grounding:** Concept 13: The edge of the map, and what's across it (Lines 823-838)

Source Grounding & Rationale: Option D is uniquely accurate per Concept 13. The text draws the exact boundary: 'Across the edge, needs more than one-prompt code: Software other people log into (accounts, payments, many users at once... those are products, not scripts)... Always-on automation with real consequences... High-stakes acts with no undo... Judgment wearing a computation costume... Inside the territory, yours as of today: every reconciliation, rollup, rename, merge, clean, cross-check, chart, and report.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because no international treaties ban AI from generating software code.
- **Option B Flaw:** Option B is incorrect because Python is the worldwide standard for quantitative finance and accounting automation.
- **Option C Flaw:** Option C is incorrect because modern AI models readily generate hundreds of lines of code in a single response.

Question 38: Correct Option A

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Concept 13: Judgment wearing a computation costume (Lines 832-833)

Source Grounding & Rationale: Option A is uniquely accurate per Concept 13. The chapter delivers this crucial ethical warning: 'Judgment wearing a computation costume. Take "compute which employees to promote" [or fire]. The arithmetic is easy. The criteria are the whole problem, and they are yours. Code computes. You decide.' Masking subjective fiduciary or human decisions as pure arithmetic abdicates non-transferable human responsibility.

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because hiding human termination criteria behind a Python script creates dangerous algorithmic bias and evades governance.
- **Option C Flaw:** Option C is incorrect because high-stakes personnel decisions must never be delegated to unsupervised autonomous agents.
- **Option D Flaw:** Option D is incorrect because productivity metrics can be quantified; the issue is who governs the termination criteria.

Question 39: Correct Option D

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Projects intro: The off-the-shelf-app test & A market of one (Lines 876-882, 932-935)

Source Grounding & Rationale: Option D is uniquely accurate per the Projects section and Project 2 ('What's My Grade, Really'). The chapter explains: 'The off-the-shelf-app test: Before you point a project at some data, ask one question. Has a company already built an app for exactly this? If yes, it is the wrong problem... The right problems are the opposite. They are a market of one, which means your rules, your mess, and an answer only you can confirm... That exact set of rules exists on one syllabus in the world, so only a brief you write can carry them.'

Distractor Dissection:

- **Option A Flaw:** Option A is incorrect because scripts running in sandboxes calculate personal grades; they do not breach institutional databases.
- **Option B Flaw:** Option B is incorrect because commercial gradebook apps routinely calculate standard weighted averages; they simply lack this specific custom replacement rule.
- **Option C Flaw:** Option C is incorrect because writing a personal grading script has nothing to do with commercial app store registrations.

Question 40: Correct Option A

Target Domain: Domain 3: AI Augmentation, Task Delegation & Agentic Workflows | **Grounding:** Projects: A catch at the end & One sentence to a real person (Lines 872-875, 910-914, 974-975)

Source Grounding & Rationale: Option A is uniquely accurate per Project 3. The text defines the ultimate goal of these projects: 'A catch is a true thing the AI's code finds that you did not already know, and that you can do something about... What you walk away with is a sentence you can say to a real person, plus a small script you keep... The deliverable is a sentence you send to the group chat tonight: "Everyone is accounted for except Usman, and there is a mystery 500, so I will message him." That sentence ends in a verb.'

Distractor Dissection:

- **Option B Flaw:** Option B is incorrect because publishing Python code to Git repository archives does not collect the missing trip dues from classmates.
- **Option C Flaw:** Option C is incorrect because a mystery memo is a routine matching question, not evidence of financial bank fraud.
- **Option D Flaw:** Option D is incorrect because creating a marketing brochure fails to resolve the personal debt reconciliation.